

Pathologic Basis of Disease: Robbins Principles

Pathology — MBBS 2nd Year

Standard Core Curriculum for Medical Undergraduates • Comprehensive Clinical Reference

Curriculum Year	MBBS 2nd Year
Medical Subject	Pathology
Target Audience	MBBS Undergraduates, Postgraduates, USMLE/PLAB Aspirants
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Chapter 1: Cellular Injury, Inflammation and Hemodynamics

Patterns of Tissue Necrosis vs Programmed Apoptosis Mechanisms

Cell death is divided into Necrosis (always pathological, with cell swelling, membrane rupture, and inflammatory response) and Apoptosis (programmed, energy-dependent cell death occurring physiologically or pathologically without inflammation).

Necrosis morphological patterns include: (1) Coagulative necrosis in ischemic solid organ infarcts (heart, kidney) where architecture is preserved for days with ghost cells lacking nuclei; (2) Liquefactive necrosis in CNS hypoxic injury (brain infarcts) and focal bacterial abscesses where enzymes rapidly digest dead cells into liquid viscous pus; (3) Caseous necrosis in tuberculous granulomas with cheesy friable debris surrounded by Langhans giant cells; (4) Enzymatic fat necrosis in pancreatitis where lipases form chalky calcium soaps.

Apoptosis occurs via the Intrinsic (Mitochondrial) and Extrinsic (Death Receptor) pathways. The intrinsic pathway is triggered by DNA damage or growth factor loss, activating BAX/BAK, releasing Cytochrome c into cytosol, forming the apoptosome with APAF-1, and activating executioner caspase-3 and caspase-7.

Clinical Pearl: Coagulative necrosis preserves cell outlines (ghost cells) in all organs except the brain, which undergoes liquefactive necrosis due to abundant hydrolytic enzymes and high lipid content.

	Classic Organs / Conditions	Key Pathogenesis
	Myocardial infarction, renal cortical infarct, splenic infarct	Denaturation of proteins
	Brain ischemic strokes, abscesses	Hydrolytic release of enzymes
	Mycobacterium tuberculosis, fungal granulomas	Delayed-type IV hypersensitivity
	Acute pancreatitis, traumatic breast injury	Pancreatic lipase

Chapter 1: Cellular Injury, Inflammation and Hemodynamics

Acute and Chronic Inflammation: Hemodynamics and Leukocyte Cascade

Acute inflammation delivers leukocytes and plasma proteins to injury sites. It comprises vascular reactions and cellular events. Vascular changes begin with transient arteriolar vasoconstriction followed by sustained vasodilation mediated by histamine and nitric oxide, producing rubor and calor.

Microvascular permeability increases through endothelial retraction induced by histamine, bradykinin, and leukotrienes, generating protein-rich exudate (specific gravity > 1.020) causing edema (tumor).

The cellular phase follows a leukocyte cascade: (1) Margination and rolling along endothelium via Selectins (L-selectin on neutrophils, E- and P-selectin on endothelium binding Sialyl-Lewis X); (2) Firm adhesion mediated by Beta-2 Integrins (LFA-1, Mac-1 binding ICAM-1); (3) Transmigration (diapedesis) via PECAM-1 (CD31); (4) Chemotaxis along gradients (C5a, LTB4, IL-8).

Clinical Pearl: Rolling is mediated by Selectins; Firm Adhesion is mediated by Integrins (CD11/CD18) binding ICAM-1. Leukocyte Adhesion Deficiency Type 1 (LAD-1) is caused by a defect in the CD18 beta-2 integrin subunit.

	Endothelial Counter-Receptor	Clinical Defect
	P-selectin (Weibel-Palade bodies), E-selectin	Leukocyte Adhesion Deficiency Type 1
	ICAM-1 (CD54), VCAM-1	Leukocyte Adhesion Deficiency Type 2
	PECAM-1 (CD31)	Transendothelial Migration Defect

Chapter 2: Cardiovascular, Renal and Neuropathology

Pathogenesis of Atherosclerosis and Histopathology of Myocardial Infarction

Atherosclerosis is a chronic inflammatory disorder of large/medium arteries initiated by endothelial injury. The 'response to injury' hypothesis details how chronic hyperlipidemia causes endothelial dysfunction, oxidized LDL accumulation in the intima, monocyte recruitment via VCAM-1, and macrophage foam cell formation.

Growth factors (PDGF, FGF) stimulate smooth muscle migration from media to intima, depositing collagen to form a fibrous cap. Plaque rupture exposes thrombogenic subendothelial collagen, triggering occlusive luminal thrombosis and acute myocardial infarction.

Histopathology of MI progresses temporally: At 0-4 hours, no light microscopic changes; at 4-12 hours, wavy fibers and early coagulation necrosis; at 12-24 hours, contraction band necrosis and hypereosinophilia; at 1-3 days, dense neutrophilic infiltrate and loss of nuclei; at 3-7 days, macrophage infiltration and dead myocyte phagocytosis (highest risk of ventricular wall rupture); at 1-2 weeks, vascular granulation tissue; at >2 months, dense collagenous fibrotic scar.

Clinical Pearl: Highest risk of myocardial rupture (ventricular free wall, septum, or papillary muscle) occurs at 3-7 days post-MI due to maximal macrophage infiltration and collagen lysis.

	Light Microscopy Hallmarks	Clinical Complications
	Minimal wavy myocardial fibers	Lethal ventricular arrhythmias
	Coagulative necrosis, contraction bands, early PMNs	Cardiogenic shock
	Loss of nuclei, heavy neutrophilic infiltrate	Fibrinous pericarditis
	Macrophage phagocytosis of necrotic debris	Free wall rupture
	Granulation tissue with neovascularization	Ventricular aneurysm

Chapter 2: Cardiovascular, Renal and Neuropathology

Neuropathology of Central Nervous System Infections: Suppurative Leptomeningitis and Viral Encephalitis

Acute Pyogenic (Suppurative) Meningitis is characterized by exudation in the leptomeninges over the cerebral hemispheres and base of the brain. Gross examination reveals engorged meningeal vessels and thick, creamy purulent exudate within the subarachnoid space (predominantly over convexity in *S. pneumoniae* and at base of brain in *H. influenzae*).

Microscopically, neutrophils fill the subarachnoid space, concentrated around leptomeningeal blood vessels. Inflammation may extend along Virchow-Robin perivascular spaces into superficial cortex, but direct parenchymal invasion is limited. Leptomeningeal phlebitis can cause venous thrombosis and hemorrhagic cerebral infarction.

In Tuberculous Meningitis, gelatinous fibrinous exudate coats the base of the brain (basilar meningitis), entrapping cranial nerves (especially CN III, IV, VI). Microscopically, caseating granulomas with Langhans giant cells and obliterative endarteritis are prominent. Viral (Aseptic) Meningitis shows mild leptomeningeal lymphocytic infiltrate without purulent exudate.

Clinical Pearl: Suppurative bacterial meningitis produces dense neutrophilic exudate in the subarachnoid space; Tuberculous meningitis characteristically causes thick gelatinous basilar exudate with cranial nerve entrapment.

	Histological Hallmark	Predominant
	Dense neutrophilic pleocytosis in subarachnoid space	Convexity (S. p
	Caseating granulomas, Langhans giant cells, endarteritis	Base of the brai
	Mild-to-moderate perivascular lymphocytic cuffing	Diffuse leptome
	Encapsulated yeast cells positive on mucicarmine and India ink	Virchow-Robin s

Chapter 2: Cardiovascular, Renal and Neuropathology

Glomerular Diseases: Nephrotic vs Nephritic Syndromes Pathophysiology

Glomerular diseases present as either Nephrotic Syndrome or Nephritic Syndrome. Nephrotic syndrome is defined by massive proteinuria (> 3.5 g/24h), hypoalbuminemia (< 3.0 g/dL), generalized edema (anasarca), and hyperlipidemia with lipiduria.

Pathogenesis involves podocyte foot process effacement destroying the negative charge barrier. Prototypes: (1) Minimal Change Disease (most common in children, normal light microscopy, podocyte effacement on electron microscopy, excellent response to corticosteroids); (2) Focal Segmental Glomerulosclerosis (FSGS, sclerosis in subsets of glomeruli, common in HIV, African ancestry); (3) Membranous Nephropathy (subepithelial immune deposits with 'spike and dome' pattern).

Nephritic syndrome features hematuria with dysmorphic RBCs and RBC casts, oliguria, azotemia, and hypertension. Prototype: Poststreptococcal Glomerulonephritis (PSGN), occurring 1-4 weeks post-streptococcal pharyngitis or impetigo, showing hypercellular glomeruli and subepithelial 'humps' on EM.

Clinical Pearl: Nephrotic syndrome = Podocyte effacement with massive proteinuria (>3.5g/day) and bland urine sediment. Nephritic syndrome = Glomerular inflammation with hematuria, dysmorphic RBCs, and RBC casts.

Nephrotic Syndrome	Nephritic Syndrome
Podocyte injury, filtration slit breakdown	Inflammatory glomerular capillary wall destruction
Massive (> 3.5 g / 24 hours)	Sub-nephrotic (< 3.5 g / 24 hours)
Absent or microscopic	Prominent (smoky / cola-colored, RBC casts)
Hypoalbuminemia reducing oncotic pressure	Renal sodium and water retention from low GFR